

MA 5104: Statistical Inference and Learning

Module 1: Statistical Models and Likelihood

Lecture 1: Why Do We Need Statistical Models?

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Why Do We Need Statistical Models?

Today's Big Question

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A Thought to Keep in Mind

"Statistics is not about numbers. It is about reasoning under uncertainty."

Case study

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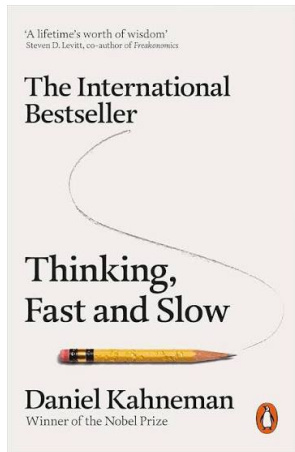
Reflect on Linda's profession

Which of the following is **more probable?**

Option A: Linda is a **bank teller.**

Option B: Linda is a **bank teller** and is **active in the feminist movement.**

Case study



Please read it if you have not!

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Why Did We Get It Wrong?

Your Brain Solved the Wrong Problem

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Philosophy; Social justice; Anti-nuclear protests; → “Sounds like Linda.”

Statistical thinking: Ignore whether the story is convincing. Ask only

$$P(A \cap B) \leq P(A)$$

Probability, not plausibility, determines the answer.

Statistics Begins with Uncertainty

Many scientific questions cannot be answered with certainty.

Examples

- Will it rain in Chennai tomorrow?
- Will a patient recover after receiving a treatment?
- Will a customer default on a loan?
- Will this machine fail within the next year?
- Will the next observation exceed 100?

Observation

The outcome is unknown *before* it is observed.
Different outcomes are possible.

Statistics is the science of reasoning under uncertainty.

Deterministic versus Stochastic Systems

Deterministic

Same input
↓
Same output

Examples

- Newton's laws
- Ohm's law
- Matrix multiplication
- Sorting algorithms

Stochastic

Same input
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Different possible outcomes

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- Disease progression
- Rainfall
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Key Insight: Statistics becomes necessary whenever variability is an intrinsic part of the system!

Where Does Variability Come From?

Suppose two patients receive

- the same treatment,
- at the same hospital,
- under the same protocol.

Where Does Variability Come From?

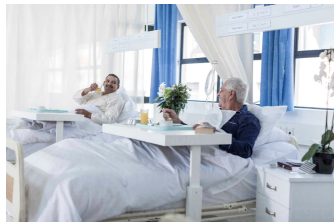
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- Biological variation
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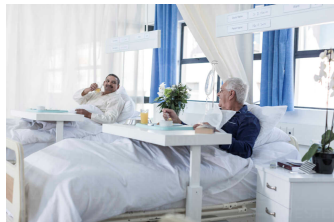
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Observation

Observed variability usually has multiple sources.

Statistics attempts to understand and quantify them.



Signal versus Noise

$$\text{Observed Data} = \text{Signal} + \text{Noise}$$

Signal

The systematic component that reflects the underlying phenomenon.

Examples: Relationship b/w dosage and recovery, Effect of fertilizer on yield

Noise

Variation that is not completely explained by the variables under study. Sources: Meas. error, Environm. fluctuations, Unobserved variables, Random chance

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Ignoring signal may cause us to overlook important scientific relationships.

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Statistical models aim to extract signal while accounting for noise.

Two Sources of Uncertainty

Aleatoric Uncertainty

Intrinsic randomness.

Cannot be eliminated by collecting more data.

Examples

- Coin tosses
- Manufacturing variability
- Weather fluctuations

Epistemic Uncertainty

Lack of knowledge.

Can often be reduced by collecting more information.

Examples

- Unknown parameters
- Missing variables
- Limited sample size

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Question

Can we distinguish these two sources of uncertainty from data alone?

Examples of Uncertainty

Problem	Aleatoric	Epistemic
Predicting patient recovery	Natural biological variation	Unknown risk factors, limited patient information
Weather forecasting	Atmospheric variability	Imperfect models, limited observations
Manufacturing quality	Process variation	Poor machine calibration, unknown process parameters

Observation

Most real statistical problems contain **both** sources of uncertainty.

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If uncertainty is epistemic

- Collect more data
- Improve the model
- Measure more variables

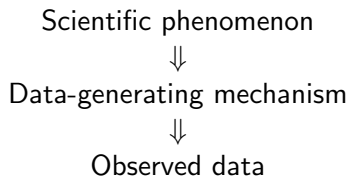
If uncertainty is aleatoric

- Estimate variability
- Quantify uncertainty
- Recognize the limits of prediction

Statistics is not about eliminating uncertainty. It is about understanding, quantifying, and managing uncertainty.

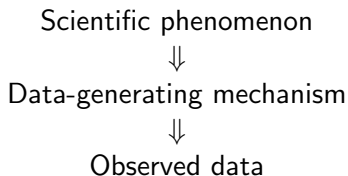
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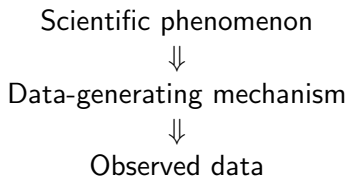
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Human physiology → Blood pressure regulation (age, genetics, diet, medication, stress, ...) → Blood pressure measurements

Consumer behavior → Purchasing decisions (preferences, income, advertising, seasonality, ...) → Customer purchases

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Statistics works **backwards**: from the observed data to the hidden mechanism.

The Law of Small Numbers

Two hospitals record births each day.

Hospital A

Approximately 45 births per day.

Hospital B

Approximately 15 births per day.

About 50% of all babies are boys.

On some days, more than 60% of the babies born are boys.

Question

Which hospital is more likely to observe such days?

- Hospital A
- Hospital B
- Both are equally likely

Our Intuition Fails Again

Most people answer

Both hospitals are equally likely

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But Randomness Behaves Differently

Smaller samples exhibit much greater variability.

Smaller sample $\implies$ Greater variability

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Law of Small Numbers

People tend to believe that small samples should closely resemble the population.

In reality, small samples fluctuate much more than large samples.

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Next Lecture

How do probability models represent uncertainty?